GH-300

GitHub Copilot

Introduction

What it is, how it works, and how to make it work well for your codebase.

What is Copilot | Architecture | Configure | Models 2026 | IDE coverage

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Deck 2 · GH-300 Certified GitHub Copilot Expert Training

OVERVIEW

What is GitHub Copilot?

An AI coding assistant that helps you write code faster and with less effort, so you can focus on problem solving and collaboration.

AI code suggestions

Autocompletes lines, blocks, and entire functions in real-time using context-aware AI.

- Whole-function generation
- Comment-aware
- Test generation support

Multi-language

Works across all major languages — from web to systems programming.

- Python · JS · TS · Go
- Ruby · C++ · Java · Rust
- HTML · CSS · SQL · Shell

Deep IDE integration

Fits natively into the tools developers already use — zero workflow disruption.

- VS Code · Visual Studio
- JetBrains family
- Neovim · Eclipse · Xcode · github.com

Feature overview

Coding assistance

- Inline suggestions
- Next Edit Suggestions
- Multi-file edits in agent
- Vision (image-to-UI)
- Model picker

Agentic workflows

- Agent · Plan · Ask · Custom
- Inline Agent
- Auto Model Selection
- Copilot cloud agent
- Copilot Spaces

Quality & review

- AI code review (CCR)
- PR summaries
- Custom instructions
- AGENTS.md
- Prompt files

Enterprise & governance

- Policy management
- EMU access control
- Audit logs
- BYOK

Configuring Copilot — github.com and VS Code

On github.com

1. Sign in → Settings → Copilot. Click Enable or start a free trial.
2. Set suggestion preferences; toggle the public-code filter.
3. Allow / block specific repositories for Copilot.
4. Org owners: Settings → Copilot → Members. Enable for all or selected.
5. Set content-exclusion rules for sensitive repos.

In VS Code

1. Extensions (Ctrl+Shift+X) → install "GitHub Copilot Chat“ if already not present.
2. Click the Copilot icon in the status bar → Sign in with GitHub.
3. Settings (Ctrl+,) → search "GitHub Copilot" → tune per-language.
4. Open Chat sidebar → pick mode (Ask · Agent · Plan · Custom).
5. Tab to accept, Esc to dismiss, Alt+] to cycle alternatives.

Copilot architecture — three components

Copilot runs in your IDE; GitHub routes requests to supported cloud-hosted models. Availability depends on plan, client and policy.

IDE extension

Runs locally in VS Code, Visual Studio, JetBrains. Captures context, sends requests, renders suggestions.

GitHub Cloud

Auth + license check, content filtering, request routing, caching, telemetry. The middleware layer.

AI models (LLMs)

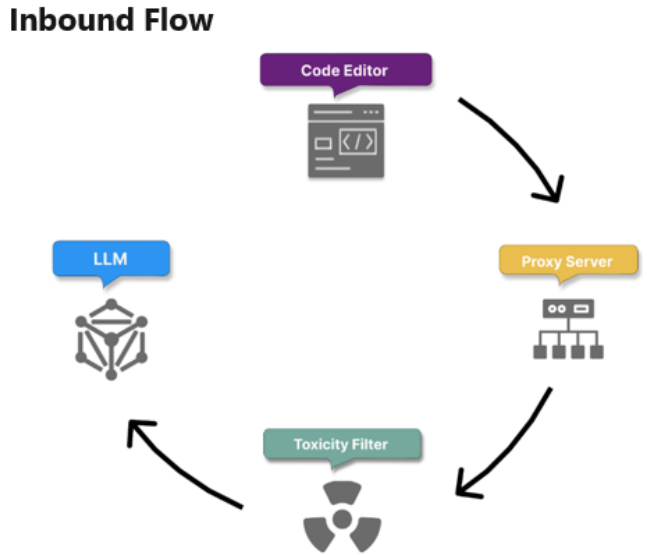
Generate code and answers.

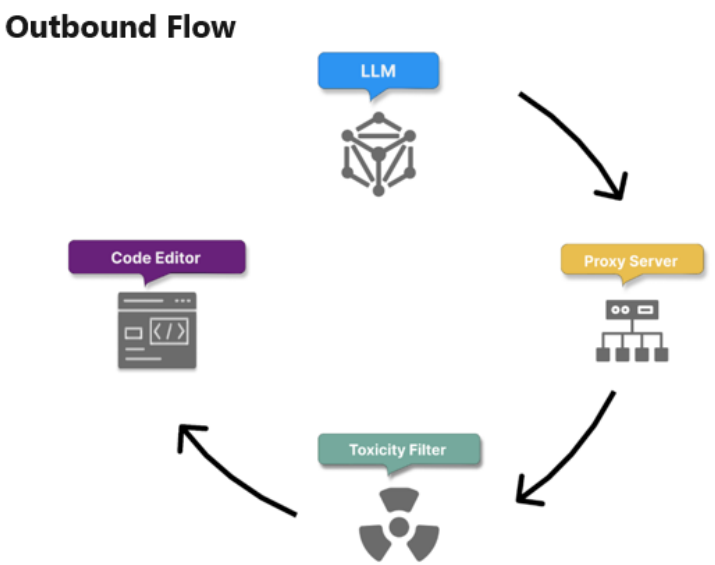
Claude: Opus 5.5, Sonnet 5.5, Haiku 5.5, Fable 5.1.

OpenAI: GPT-6.1 Sol; GPT-6 Astra; GPT-6 Luna; GPT-5.6 Terra.

Google: Gemini 3.8 Flash.

Copilot architecture — Inbount and Outbound Flow





Code quality = suggestion quality

Poor naming	Good naming	What Copilot understands
`getData(x)`	`getUserById(userId)`	Fetches a user by ID
`calc(a)`	`calculateTaxWithDiscount(amt)`	Calculates tax + applies discount
`check(str)`	`validateEmailFormat(email)`	Validates email format
`process(x)`	`validate_email_address(email: str)`	Generates regex validation logic

Other quality signals Copilot picks up on

- Consistent patterns — if your first 5 functions follow a pattern, Copilot continues it for function 6.
- Type annotations — TypeScript and typed languages give Copilot more semantic information.
- Existing tests — Arrange-Act-Assert patterns in your suite drive matching test generation.
- Targeted comments — "Calculate total with 15% discount for orders over $100" specifies intent.

Latest models — pick by task, not by habit

Model	Strength	Example prompts	USD / 1M tokens
Claude Fable 5.1	Long-horizon autonomous work; fewer tool calls	1. Upgrade every service to .NET 9 2. Trace how auth works across repo	In $10 · Out $50
Claude Opus 5.5	Long-running agents; strong error recovery	1. Split OrderService into modules 2. Find why checkout tests flake	In $4 · Out $20
Claude Sonnet 5.5	Efficient coding; fewer tool calls	1. Add CSV export with unit tests 2. Rename userId everywhere	In $2 · Out $10
GPT-6 Astra	Plans, validates and self-checks its work	1. Build signup flow and test it 2. Test checkout flow in browser	In $10 · Out $50
GPT-6.1 Sol	Advanced reasoning; efficient complex coding	1. Find this memory leak's cause 2. Assess our schema for scaling	In $2 · Out $10
GPT-5.6 Terra	Balanced interactive and agentic coding	1. Add pagination to /orders API 2. Review this PR for bugs	In $2 · Out $12
GPT-6 Luna	Low-cost, responsive lightweight assistance	1. Fix null check in login.ts 2. Add docstrings to utils.py	In $0.10 · Out $0.50
Gemini 3.8 Flash	Fast assistance for lightweight coding	1. Explain what this regex does 2. Mock up a todo app in React	In $0.75 · Out $3.75

Source: GitHub Docs & Changelog · model support & pricing (default tier, uncached)

Thank you.

Open Copilot Chat. Pick a mode. Build something small.

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